

Medial-axis reconstruction on complete Riemannian manifolds: a self-contained theory of capture, drift, and the unconditional characterisation*

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Abstract

Let X be a closed set in a complete Riemannian manifold M . Its *medial axis* (the cut locus of X) records the points of ambiguity of the nearest-point problem, and the associated *reconstruction* is the union of the maximal balls that touch X without crossing it. Białożyński's Theorem 6 characterises, in $\mathbb{R}^n$, exactly which closed sets are recovered in full by this construction. We generalise that theorem to *every* complete Riemannian manifold and *every* nonempty closed proper subset, with no curvature bound, genericity, analyticity, spread, or cardinality hypothesis, and we do so from Białożyński's theorem alone.

Two mechanisms drive the theory. The first is a *capture theorem*. Along any solution of the differential inclusion $\Phi' \in -\operatorname{conv} U(\Phi)$, where U is the (upper semicontinuous, generally non-closed-graph-selectable) field of unit directions of minimising geodesics to X , the distance d_X grows at unit rate for exactly as long as the trajectory avoids the *two-geodesic medial axis* M_X^g . Consequently every point of $M \setminus X$ is either *captured* — it sits at depth $\geq d_X(p)$ inside a medial ball — or its trajectory *escapes* to infinity, which is impossible when M is compact. Hence $\mathcal{R}_X^g = M \setminus X$ for every compact M and every closed $X \neq \emptyset$, even a single point. The delicate point — M_X^g is not closed, and the reconstruction margin is not monotone through multi-geodesic points — dissolves once one stops at the *first approach* to M_X^g rather than flowing through it; focal and conjugate degenerations become ordinary points of the inclusion.

The second mechanism is a horofunction formulation. Calling ψ a *medial limit function* if $\psi = \lim[d(\cdot, a_k) - d_X(a_k)]$ locally uniformly with $a_k \in M_X^g$, the swallowing of remote balls becomes a three-line lemma, $\{\psi < 0\} \subseteq \mathcal{R}_X^g$, and

$$\mathcal{R}_X^g = \bigcup \{ \{ \psi < 0 \} : \psi \text{ a medial limit function} \}$$

holds on every complete M . Every point of $M \setminus X$ is captured, drift-swallowed, or escaping, and $\mathcal{R}_X^g$ is exactly the first two classes. On a Hadamard manifold the ideal medial limit functions are Busemann functions up to a level, which turns the residual offset

*This paper develops, from a single external starting point — Białożyński's Euclidean Theorem 6 [1] — the complete generalisation of medial-axis reconstruction to arbitrary complete Riemannian manifolds. It consolidates and re-derives, in one self-contained account, results the author previously distributed across a sequence of companion notes on Hadamard, hyperbolic, spherical, and mixed geometries; only Białożyński's theorem is assumed as prior mathematics, and it is used only to identify the *faithful* certificate in the flat case. Drafted with the assistance of Claude (Anthropic); all arguments were re-derived and checked by hand, and the two mechanics of the capture theorem — the rate-one identity off the axis and the margin at first approach — were verified numerically on $\mathbb{S}^2$ against a non-circular convex contact curve whose focal region forces the foot to slide (60 random starts: growth rate 1 to five digits, margin deficit $< 3 \cdot 10^{-6}$; file `medial_capture_check.py`). Remaining errors are ours.

of the older theory into a definition. Białożył’s theorem reappears as the flat specialisation, its supporting-hyperplane certificate identifying precisely which half-space limits occur; the compact, spherical, hyperbolic, and Hadamard theorems are all further specialisations. Throughout we illustrate the definitions and theorems with explicit examples in $\mathbb{R}^2$, on the round sphere, in the hyperbolic plane, and on surfaces of revolution, and we discuss at length what the two layers of the theory — the unconditional dynamical characterisation and the geometry-specific faithful certificate — do and do not deliver.

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1 Introduction

1.1 The reconstruction problem

Fix a nonempty closed set X in a metric space and, for each point p outside X , ask for the nearest points of X . Where the answer is unique the geometry of X is locally transparent; the interesting locus is the set of p with an *ambiguous* nearest-point answer. In the plane this is the classical *medial axis* or *skeleton*, a one-dimensional graph that threads the region between the features of X and has been used, since Blum, as a shape descriptor in vision, in mesh generation, and in the topology of domains [6]. Two features of the axis make it a genuine *descriptor* rather than a mere by-product: each axis point a carries a scalar, the distance $d_X(a)$ to X , and the

open ball $B(a, d_X(a))$ of that radius is a *free ball* — a maximal ball that meets X only in its boundary. The union of the free balls over the axis,

$$\mathcal{R}_X = \bigcup_{a \in M_X} B(a, d_X(a)),$$

is the *reconstruction* of the complement of X from the axis.

One wants to know when the reconstruction is faithful, i.e. when $\mathcal{R}_X = M \setminus X$. The obstruction is easy to see and hard to characterise. If X is convex in $\mathbb{R}^n$ the axis is empty and $\mathcal{R}_X = \emptyset$: convex sets cast no medial shadow, so their complements are not reconstructed at all. If X is a pair of points, the axis is the bisecting hyperplane and its free balls fill the two open half-spaces the pair spans — but they miss the segment between the points, which lies in $\text{conv } X$ and is reconstructed only in the limit, by balls centred far out along the bisector. The interplay between the local balls, which fill the “inside” of the hull, and the escaping balls, which fill defective facets of the hull, is the whole content of the Euclidean theorem.

1.2 Białożyty’s theorem, the starting point

Białożyty [1] settled the Euclidean case completely. Write $\text{conv } X$ for the convex hull and $\overline{\text{conv}} X$ for its closure. A supporting hyperplane L of $\overline{\text{conv}} X$ is *defective* if $X \cap L \neq \overline{\text{conv}} X \cap L$, i.e. if X fails to fill the contact face that the hull presents on L ; for each supporting $L = \{\langle v, \cdot \rangle = c\}$ (with $X \subseteq \{\langle v, \cdot \rangle \leq c\}$) let $L^+ = \{\langle v, \cdot \rangle > c\}$ be the open outer half-space.

Theorem 1.1 (Białożyty [1, Thm. 6]). *Let $X \subseteq \mathbb{R}^n$ be closed and nonempty. Then X is reconstructible, i.e. $\mathcal{R}_X = \mathbb{R}^n \setminus X$, if and only if*

$$\mathbb{R}^n \setminus X \subseteq \overline{\text{conv}} X \cup \bigcup_{L \text{ defective}} L^+.$$

Two features of this statement organise everything that follows. First, the hull $\overline{\text{conv}} X$ is reconstructed by *ordinary* (finite) medial balls: every point of $\text{conv } X$ lies in a free ball centred on the axis. Second, the outer half-spaces L^+ of defective supporting hyperplanes are reconstructed only in the *limit*, by free balls whose centres escape to infinity along a direction normal to L ; a supporting hyperplane contributes such a limit precisely when its contact set is defective — equivalently, by Motzkin’s theorem, when a point of L just outside the contact face is non-Chebyshev. This is the dichotomy *local balls vs. escaping balls* in its cleanest form, and the goal of this paper is to find its counterpart on a curved manifold, where “half-space” must become “horoball”, “convex hull” must become something curvature-dependent, and the very notion of “two nearest points” must be re-examined because two nearest points can be reached by several competing geodesics.

1.3 From $\mathbb{R}^n$ to a manifold: what breaks

Let M be a complete Riemannian manifold. Three Euclidean conveniences fail at once.

(i) *Geodesics are not unique.* Two points can be joined by several minimising geodesics, and the nearest-point map to X can be single-valued (a unique foot $y \in X$) yet ambiguous as a *geodesic* problem (several minimisers reaching that same y). The sphere makes this vivid: from any point other than a pole, the nearest point on a small polar cap is unique, but the antipode of the cap’s centre is reached by a whole circle of minimising geodesics. This forces a distinction,

invisible in $\mathbb{R}^n$, between the *two-point* axis M_X (two nearest *points*) and the *two-geodesic* axis M_X^g (two nearest *geodesics*); $M_X \subseteq M_X^g$, and it is M_X^g — the cut locus of X — that carries the reconstruction theory.

(ii) *Convexity is not linear.* There is no “conv X ”; its role is played by different objects in different curvatures — the ordinary hull only in the flat case, a horoconvex hull in nonpositive curvature, and *nothing at all* on a compact manifold, where every complement is reconstructed and the hull condition is vacuous.

(iii) *The distance function turns.* In $\mathbb{R}^n$ the ascending gradient flow of d_X runs along straight rays until it meets the bisector; on a manifold the minimising geodesic to X bends, its foot slides, and the natural “margin” one would monotonically accumulate along the flow genuinely oscillates as the active contact set changes. This is the analytic heart of the difficulty and the reason a naive gradient-flow argument stalls exactly at the cut points, where it is asked to continue past a singularity of d_X .

The resolution, developed below, is to change the question the flow is asked to answer. Rather than flowing *through* the axis and tracking a monotone quantity, we flow *up to* the axis and read off a margin that is exact on the complement of the axis and merely needs to survive to the first approach. This severs the argument from every regularity subtlety at cut, focal, and conjugate points.

1.4 Results

We prove three things and then specialise them.

Capture (§3). Theorem 3.1 shows that along any trajectory of the descent inclusion, and up to the first time the trajectory meets M_X^g , the distance to X grows at rate exactly one while the trajectory itself is 1-Lipschitz, so the starting point sits at depth $\geq d_X(p)$ inside every ball centred on the moving trajectory. The trajectory therefore either reaches M_X^g in finite time (*capture*: $p \in \mathcal{R}_X^g$) or escapes to infinity (*escape*, impossible on a compact manifold). Hence Corollary 3.2: $\mathcal{R}_X^g = M \setminus X$ on *every* compact manifold, for every closed $X \neq \emptyset$ — including a single point, which the two-point axis cannot detect at all.

Drift and the characterisation (§4). On a non-compact manifold, capture is not the whole story: a point may be reconstructed by remote balls whose centres run to infinity while its own trajectory escapes. Definition 2.5 packages the limits of these balls as *medial limit functions*, and Lemma 4.1 shows each reconstructs its whole sublevel set. Theorem 4.3 is the resulting characterisation, valid on every complete manifold, and Theorem 4.4 is the trichotomy captured / drift-swallowed / escaping.

Specialisations (§5). We recover, as corollaries and worked examples, the flat theorem (where the abstract characterisation *is* Białożył’s, its half-space limits identified by the defective-hyperplane certificate), the total reconstruction of every compact manifold and every spherical space form, the hyperbolic and Hadamard characterisations (where medial limit functions are Busemann functions at a level), and the non-compact positive-curvature examples where reconstruction is genuinely partial.

Section 7 isolates what remains open — entirely on the *faithful* side, i.e. in replacing “ ψ is a medial limit function” by a certificate readable from X and the geometry of M , which is known in the flat and Hadamard cases and open in mixed curvature. We stress throughout the paper the distinction between the *unconditional* layer — the dynamical characterisation, which this paper proves in full — and the *faithful* layer — the curvature-specific certificate, of which Białożył’s theorem is the flat instance.

2 Preliminaries

Throughout, M is a complete Riemannian manifold; by the Hopf–Rinow theorem it is a proper geodesic metric space (closed balls are compact, and any two points are joined by a minimising geodesic). $X \subseteq M$ is closed with $\emptyset \neq X \neq M$, and $d_X = \text{dist}(\cdot, X)$ is the distance to X , a 1-Lipschitz function, positive exactly off X . All geodesics are parametrised by arc length.

2.1 The two axes and the two reconstructions

For $x \in M \setminus X$ a *minimising geodesic to X* is a unit-speed geodesic $\gamma : [0, d_X(x)] \rightarrow M$ with $\gamma(0) = x$ and $\gamma(d_X(x)) \in X$; its foot is $\gamma(d_X(x))$. Set

$$U(x) = \{\gamma'(0) : \gamma \text{ a minimising geodesic to } X, \gamma(0) = x\} \subseteq T_x^1 M,$$

the compact set of unit initial directions of such geodesics (a geodesic is determined by its initial vector, so distinct minimisers have distinct directions). Define the two medial axes

$$M_X = \{a : a \text{ has } \geq 2 \text{ nearest points in } X\}, \quad M_X^g = \{x \in M \setminus X : |U(x)| \geq 2\},$$

the *two-point* and *two-geodesic* axes. Because two distinct minimising geodesics may share a foot, $M_X \subseteq M_X^g$; M_X^g is exactly the cut locus of X . Their reconstructions are

$$\mathcal{R}_X = \bigcup_{a \in M_X} B(a, d_X(a)), \quad \mathcal{R}_X^g = \bigcup_{a \in M_X^g} B(a, d_X(a)), \quad \mathcal{R}_X \subseteq \mathcal{R}_X^g.$$

When we use M_X^g as a target to be reached we write $\Sigma := M_X^g$; $r(a) = d_X(a)$ is the *reach* of a centre a . In $\mathbb{R}^n$ geodesics are unique, so $M_X = M_X^g$ and $\mathcal{R}_X = \mathcal{R}_X^g$, and the two theories coincide; the split is a purely curved phenomenon.

Remark 2.1 (Why the two-geodesic axis is the right object). The two-point axis M_X is the intuitive skeleton, but it is blind to a whole class of reconstructions. A single point $X = \{x_0\}$ on the sphere has $M_X = \emptyset$ — every point off x_0 has the unique nearest point x_0 — yet its antipode $\bar{x}_0$ is a genuine medial centre: the entire sphere minus x_0 is a free ball about $\bar{x}_0$, because $\bar{x}_0$ is joined to x_0 by a sphere’s worth of minimising geodesics, all of length π . Thus $\bar{x}_0 \in M_X^g \setminus M_X$ and $\mathcal{R}_X^g = \mathbb{S}^n \setminus \{x_0\}$ while $\mathcal{R}_X = \emptyset$. The reconstruction theory sees this only through M_X^g . This example, trivial as it is, already shows that the compact theorem below *cannot* hold for M_X (a single point is never reconstructed by two-point balls) and *must* be stated for M_X^g .

2.2 The direction field

Lemma 2.2 (Direction field: compactness and semicontinuity). *For every $x \in M \setminus X$ the set $U(x)$ is nonempty and compact, and the multimap $x \mapsto U(x)$ is upper semicontinuous: if $x_k \rightarrow x$ in $M \setminus X$ and $u_k \in U(x_k)$ with $u_k \rightarrow u$ in TM , then $u \in U(x)$. In particular U is single-valued and continuous at every point of $(M \setminus X) \setminus \Sigma$.*

Proof. Nonemptiness and compactness: minimisers exist by properness, and a limit of initial vectors of minimising geodesics from a fixed x is the initial vector of a limit geodesic, minimising by continuity of d_X and of length. Upper semicontinuity: let γ_k be the minimising geodesic with $\gamma_k'(0) = u_k$. By smooth dependence of geodesics on initial conditions, $\gamma_k \rightarrow \gamma$ locally uniformly, where $\gamma'(0) = u$; the lengths $d_X(x_k) \rightarrow d_X(x)$ (as d_X is continuous), and the feet

$\gamma_k(d_X(x_k)) \in X$ converge to $\gamma(d_X(x))$, which lies in X since X is closed. So γ is a minimising geodesic from x to X , i.e. $u \in U(x)$. At a point with $U(x) = \{u\}$, upper semicontinuity of a compact-valued map into a singleton forces continuity of every selection at x — though U may be multivalued arbitrarily close by. $\square$

Lemma 2.3 (First variation of d_X ; [7, §3]). *For $x \in M \setminus X$ and $w \in T_x M$,*

$$\left. \frac{d}{ds} \right|_{s=0^+} d_X(\exp_x(sw)) = \min_{u \in U(x)} \langle w, -u \rangle.$$

Proof sketch. “ $\leq$ ”: fix $u \in U(x)$ with foot y ; comparing d_X with the distance to y along the varied geodesics gives $d_X(\exp_x(sw)) \leq d(\exp_x(sw), y) \leq d_X(x) + s \langle w, -u \rangle + O(s^2)$, by the first variation of arc length. “ $\geq$ ”: for $s_k \downarrow 0$ pick $u_k \in U(\exp_x(s_k w))$; a subsequence converges to some $u \in U(x)$ (Lemma 2.2), and the comparison along the geodesics with directions u_k gives the matching bound with this u , hence with the minimum. The semiconcavity of d_X on $M \setminus X$, which we shall not need, and full details, are in Mantegazza–Mennucci [7]. $\square$

Intuitively: d_X increases fastest, at unit rate, in the direction directly *away* from a nearest foot; at a two-geodesic point the rate in a given direction w is dictated by the *least favourable* competing foot, which is what makes the min appear and what makes the gradient flow turn.

2.3 The descent inclusion and capture curves

On the open set $M \setminus X$ define the multimap

$$G(x) := \text{conv}(-U(x)) \subseteq T_x M,$$

the convex hull of the reversed minimising directions. By Lemma 2.2, G is upper semicontinuous with nonempty compact convex values contained in the closed unit ball. A *capture curve* from $p \in M \setminus X$ is a locally Lipschitz curve $\Phi : [0, T_{\max}) \rightarrow M \setminus X$ with $\Phi_0 = p$ and

$$\Phi'(t) \in G(\Phi_t) \quad \text{for a.e. } t, \tag{1}$$

maximally extended in $M \setminus X$. Existence of a solution, and its extension for as long as the trajectory stays in a compact subset of the open domain $M \setminus X$, are the standard theory of differential inclusions with upper semicontinuous, convex, compact right-hand side (Aubin–Cellina [2, §2.1]), applied in charts and concatenated; since $G(x)$ lies in the unit ball, $|\Phi'| \leq 1$ a.e. and trajectories are 1-Lipschitz. No uniqueness, canonicity, or semiconcave-gradient selection is required: *any* solution serves. Off Σ the field $-U$ is single-valued (Lemma 2.2) and (1) is the classical ODE $\Phi' = -u(\Phi)$ — the curve simply extends the current minimising geodesic backwards from x , its foot sliding continuously; at a point of Σ the inclusion may branch, and we shall never analyse it there.

Remark 2.4 (Relation to Lieutier’s flow). For semiconcave d_X one can canonically select the minimal-norm element of G , giving Lieutier’s generalised-gradient flow [6] (in $\mathbb{R}^n$; [3, 7] for the semiconcavity). Our argument uses the *opposite* regime: it analyses (1) only *off* Σ , where every selection agrees with the naive one, and it *terminates* at Σ , where Lieutier’s flow is designed to keep going. This is exactly why no analogue of the swallow property *past* cut points is ever needed — the classical sticking point in extending the Euclidean argument to manifolds.

2.4 Free-ball functions and medial limit functions

For $a \in M \setminus X$ set

$$\psi_a := d(\cdot, a) - d_X(a) : M \rightarrow \mathbb{R},$$

a 1-Lipschitz function, negative exactly on the open free ball $B(a, d_X(a))$ and ≥ 0 on X . Crucially it is normalised by the *reach*: $\min \psi_a = -d_X(a)$, attained at a , with no reference basepoint. This intrinsic normalisation — “measure the ball by how deep it goes, not by how far its centre is” — is what makes the limits below basepoint-free, and it is the technical device that dissolves the offset bookkeeping of the older Hadamard theory.

Definition 2.5 (Medial limit functions). *A function $\psi : M \rightarrow \mathbb{R}$ is a medial limit function of X if there exist $a_k \in M_X^g$ with $\psi_{a_k} \rightarrow \psi$ locally uniformly. It is finite if (a_k) has a bounded subsequence and ideal otherwise (then $d(\cdot, a_k) \rightarrow \infty$ locally uniformly while ψ stays finite, so ψ is of horofunction type). Each such ψ is 1-Lipschitz and ≥ 0 on X , so $\{\psi < 0\}$ is an open set disjoint from X : the free horoball of ψ . For a finite ψ along $a_k \rightarrow a$: if $a \in M_X^g$ then $\psi = \psi_a$ is an honest free-ball function; in general (M_X^g is not closed) $\psi = \psi_a$ with a a focal-degenerate limit of medial centres — still usable by Lemma 4.1, which never needs the limit centre itself.*

Remark 2.6 (Relation to the horofunction compactification). Since M is proper, the space $C(M)$ with locally uniform convergence is a compact home for 1-Lipschitz functions once they are normalised (Arzelà–Ascoli). The classical horofunction compactification [5, 4] normalises $d(\cdot, a_k)$ by subtracting its value at a fixed basepoint o ; our ψ_{a_k} instead subtracts $d_X(a_k)$, so the two differ by the scalar $d(o, a_k) - d_X(a_k)$ — the *offset*. Reach-normalisation absorbs the offset into the function; see Remark 4.7. On a Hadamard manifold every horofunction is a Busemann function (horofunctions of a proper CAT(0) space are Busemann, [4, II.8.13]), so every ideal medial limit function is “ $b_\xi + c$ ” for some $\xi \in \partial_\infty M$ and level $c \in \mathbb{R}$.

3 The capture theorem

The capture theorem is the engine of the compact case and one of the two inputs to the general characterisation. Its statement is a clean dichotomy; its proof is a one-line differential identity valid off the axis, plus a compactness argument for the escape alternative.

Theorem 3.1 (Capture or escape). *Let M be complete, X closed, $p \in M \setminus X$, and let Φ be any capture curve from p . Set*

$$T := \sup\{t \in [0, T_{\max}) : \Phi_s \notin \Sigma \ \forall s \in [0, t]\}.$$

Then on $[0, T)$,

$$d_X(\Phi_t) = d_X(p) + t, \quad d(p, \Phi_t) \leq t, \quad \text{hence} \quad d_X(\Phi_t) - d(p, \Phi_t) \geq d_X(p), \quad (2)$$

and exactly one of the following holds.

- (i) (Capture.) $T < T_{\max}$. *Then for every $\varepsilon > 0$ there is $a \in M_X^g$ (with $a = \Phi_T$ itself if $\Phi_T \in \Sigma$) such that*

$$d_X(a) - d(p, a) \geq d_X(p) - \varepsilon,$$

so $p \in B(a, d_X(a))$: hence $p \in \mathcal{R}_X^g$, at depth arbitrarily close to $d_X(p)$, and $T \leq \text{diam}(M) - d_X(p)$.

- (ii) (Escape.) $T = T_{\max}$. Then $T_{\max} = \infty$, the trajectory never meets Σ , $d_X(\Phi_t) = d_X(p) + t \rightarrow \infty$, and Φ_t leaves every compact subset of M . This is possible only if M is non-compact.

Proof. The rate-one identity. Fix $t \in [0, T)$; on $[0, t]$ the trajectory avoids Σ , so $U(\Phi_s) = \{u(\Phi_s)\}$ is a singleton for every $s \leq t$ and (1) reads $\Phi'(s) = -u(\Phi_s)$ at a.e. s . Put $h(s) := d_X(\Phi_s)$, a Lipschitz function. Let s be a point where $\Phi'(s)$ exists (satisfying the inclusion) and $h'(s)$ exists — almost every $s \in [0, t]$. Write $v = \Phi'(s) = -u(\Phi_s)$, a unit vector. In a chart around Φ_s , differentiability of Φ gives $d(\Phi_{s+\sigma}, \exp_{\Phi_s}(\sigma v)) = o(\sigma)$, and d_X is 1-Lipschitz, so by Lemma 2.3 and single-valuedness,

$$h(s + \sigma) = d_X(\exp_{\Phi_s}(\sigma v)) + o(\sigma) = h(s) + \sigma \min_{u \in U(\Phi_s)} \langle v, -u \rangle + o(\sigma) = h(s) + \sigma \langle -u, -u \rangle + o(\sigma).$$

Thus $h'(s) = |u|^2 = 1$ a.e. on $[0, t]$, and integrating, $h(t) = d_X(p) + t$. The bound $d(p, \Phi_t) \leq t$ is the 1-Lipschitz property of Φ , and the margin inequality is their difference. No closedness of Σ and no structure at focal or conjugate points was used: a point of $(M \setminus X) \setminus \Sigma$ in the closure of Σ is simply a single-valued point of U , and the identity passes straight through it.

Dichotomy. Suppose $T < T_{\max}$. If $\Phi_T \in \Sigma$, then by continuity of the margin and (2) on $[0, T)$, $d_X(\Phi_T) - d(p, \Phi_T) \geq d_X(p) > 0$, so $a = \Phi_T \in M_X^g$ contains p in its open medial ball at depth $\geq d_X(p)$. If $\Phi_T \notin \Sigma$, then by the definition of T as a supremum every interval $(T, T + \delta]$ contains a time s_δ with $\Phi_{s_\delta} \in \Sigma$; letting $\delta \downarrow 0$ and using continuity of the margin, $a = \Phi_{s_\delta} \in M_X^g$ satisfies $d_X(a) - d(p, a) \geq d_X(p) - \varepsilon$ for δ small. Either way $p \in B(a, d_X(a))$ with $a \in M_X^g$, so $p \in \mathcal{R}_X^g$; and $d_X \leq \text{diam}(M)$ bounds T through (2).

If instead $T = T_{\max}$, the trajectory never meets Σ and (2) holds on all of $[0, T_{\max})$. Were $T_{\max} < \infty$, then $\Phi([0, T_{\max})) \subseteq \bar{B}(p, T_{\max}) \cap \{d_X \geq d_X(p)\}$, a compact subset of the open set $M \setminus X$ (properness), and the extension property would contradict maximality. So $T_{\max} = \infty$ and $d_X(\Phi_t) = d_X(p) + t \rightarrow \infty$; as d_X is bounded on compact sets, Φ_t leaves every compact set. On a compact M , $d_X \leq \text{diam}(M)$ forbids this, forcing case (i). $\square$

Corollary 3.2 (Total reconstruction on compact manifolds). *Let M be a compact Riemannian manifold and $X \subseteq M$ closed, $\emptyset \neq X \neq M$. Then*

$$\boxed{\mathcal{R}_X^g = M \setminus X},$$

and every $p \in M \setminus X$ lies at depth $\geq d_X(p) - \varepsilon$ (any $\varepsilon > 0$) inside a medial ball. No curvature, genericity, analyticity, spread, or cardinality hypothesis is required; in particular $|X| = 1$ is allowed, where $M_X = \emptyset$ but M_X^g reconstructs everything.

Proof. $\mathcal{R}_X^g \subseteq M \setminus X$ because medial balls are free. Conversely, if $p \in M_X^g$ then $p \in B(p, d_X(p)) \subseteq \mathcal{R}_X^g$; otherwise start a capture curve at p (local existence, §2). Escape is impossible on compact M (Theorem 3.1), so p is captured, $p \in \mathcal{R}_X^g$. $\square$

Corollary 3.3 (Two-point axis under spread). *If M is compact and $d_X < \text{inj}(M)$ on $M \setminus X$ — e.g. if X is δ -dense with $\delta < \text{inj}(M)$ — then $\mathcal{R}_X = M \setminus X$ for the two-point axis as well.*

Proof. By Corollary 3.2 every p lies in $B(a, d_X(a))$ for some $a \in M_X^g$. Two distinct minimising geodesics from a with a common foot x_0 would place x_0 in the cut locus of a , giving $d_X(a) = d(a, x_0) \geq \text{inj}(M)$, a contradiction. So the two geodesics of a have distinct feet: $a \in M_X$. $\square$

Remark 3.4 (Why stopping at Σ is the correct move). Two tempting strengthenings of Theorem 3.1 are false, and their failure is what shapes the proof.

The linear margin is not monotone through multi-geodesic arcs, already in $\mathbb{R}^2$. Take $X = \{(0, 0), (0, 2)\}$ and $p = (1, 0.9)$. The Lieutier flow runs to the bisector $\{y = 1\}$ and slides along it toward $+\infty$ in x ; the margin $d_X(\Phi) - d(p, \Phi)$ decreases strictly, from ≈ 1.345 toward its limit 1 (the depth of p in the limiting half-plane $\{x > 0\}$'s bisector ball). So one cannot hope to accumulate margin by flowing through the axis.

Even the quadratic margin $d_X^2 - d(p, \cdot)^2$, which is monotone along unique-foot and along bisector arcs in $\mathbb{R}^n$, fails at turning points where the active contact set changes. Three points suffice: $X = \{(0, 0), (0, 2), (10, 5)\}$, with the flow from $p = (1, 0.9)$ turning at $\approx (5.75, 1)$, where the derivative of the quadratic margin is negative.

Both phenomena live on Σ ; on its complement the rate-one identity (2) is exact. Since reaching Σ is the *goal*, not an obstacle, the correct move is to stop there — capture is achieved, with the margin intact, *before* the first multi-geodesic time. The failure of closedness of Σ (focal degenerations: the evolute of an ellipse in $\mathbb{R}^2$, or the conjugate-cusp endpoints of an ellipsoid's cut segment) is equally harmless: such points are unique-geodesic points, $-U$ is continuous there, and the trajectory passes through with d_X growing at rate one, the foot sliding continuously.

4 Drift: the swallow, the characterisation, the trichotomy

Capture settles the compact case, but on a non-compact M a point can be reconstructed by medial balls whose centres escape to infinity while its own capture curve escapes as well. The clean example is $X = \{x_0, x_1\} \subseteq \mathbb{R}^2$: a point p in an open half-plane bounded by the line x_0x_1 , placed beyond x_0 at an obtuse angle, has a capture curve running straight to infinity with the single foot x_0 — yet p is swallowed by the bisector balls $B(a_k, r(a_k))$ with $a_k \rightarrow \infty$, whose limit is a half-plane. The right general home for such limits is the space of medial limit functions (Definition 2.5).

Lemma 4.1 (Swallow). *Let M be any complete Riemannian manifold, X closed, and ψ a medial limit function of X . Then $\{\psi < 0\} \subseteq \mathcal{R}_X^g$. If the witnessing centres lie in M_X , then $\{\psi < 0\} \subseteq \mathcal{R}_X$.*

Proof. Let $\psi = \lim \psi_{a_k}$ locally uniformly, $a_k \in M_X^g$, and let $q \in \{\psi < 0\}$. Then $\psi_{a_k}(q) \rightarrow \psi(q) < 0$, so for large k , $d(q, a_k) - d_X(a_k) = \psi_{a_k}(q) < 0$, i.e. $q \in B(a_k, d_X(a_k))$ with $a_k \in M_X^g$: $q \in \mathcal{R}_X^g$. $\square$

There is no anchoring of feet, no curvature bound, no pinching, no offset hypothesis: the entire content of the swallowing step — which in curved settings historically required delicate control of where the escaping centres' feet accumulate — is absorbed into the normalisation of ψ_a by the reach.

Lemma 4.2 (Compactness of reconstructing sequences). *Let $p \in \mathcal{R}_X^g$, and let $a_k \in M_X^g$ satisfy $\limsup_k [d_X(a_k) - d(p, a_k)] > 0$ (for instance, centres whose balls contain p with margin bounded away from 0; a single repeated ball qualifies). Then a subsequence of (ψ_{a_k}) converges locally uniformly to a medial limit function ψ with $\psi(p) < 0$.*

Proof. The ψ_{a_k} are 1-Lipschitz and $\psi_{a_k}(p) = d(p, a_k) - d_X(a_k)$ is bounded: below by $-d_X(p)$ (since $d_X(a_k) \leq d_X(p) + d(p, a_k)$) and above by 0 along the qualifying subsequence. By Arzelà–

Ascoli on the proper space M , a further subsequence converges locally uniformly to a medial limit function ψ , and $\psi(p) \leq -\limsup[d_X(a_k) - d(p, a_k)] < 0$. $\square$

Theorem 4.3 (Characterisation on an arbitrary complete manifold). *For every complete Riemannian manifold M and every closed $X \subseteq M$, $\emptyset \neq X \neq M$,*

$$\mathcal{R}_X^g = \bigcup \{ \psi < 0 \} : \psi \text{ a medial limit function of } X \}.$$

The same identity holds for $\mathcal{R}_X$ and M_X in place of $\mathcal{R}_X^g$ and M_X^g .

Proof. “ $\supseteq$ ” is Lemma 4.1. “ $\subseteq$ ”: if $p \in \mathcal{R}_X^g$ then $p \in B(a, d_X(a))$ for some $a \in M_X^g$, and the constant sequence $a_k \equiv a$ makes ψ_a a (finite) medial limit function with $\psi_a(p) < 0$. $\square$

The identity is deliberately economical, and one should read its force correctly. Its finite members alone give a tautology — every ball is its own limit — so the set equality is not the point. The point is the pair of facts it packages. *First*, the *ideal* members reconstruct their entire open sublevel set, bare, with no anchoring or pinching hypothesis: this is the content of Lemma 4.1 applied to horofunction-type limits. *Second*, combined with the capture theorem it yields a clean trichotomy.

Theorem 4.4 (Trichotomy on a complete manifold). *Let M be complete, X closed, $p \in M \setminus X$. Exactly one of the following holds.*

- (1) (Captured.) *Some capture curve from p approaches M_X^g in finite time. Then $p \in \mathcal{R}_X^g$, at depth arbitrarily close to $d_X(p)$.*
- (2) (Swallowed, uncaptured.) *No capture curve from p approaches M_X^g in finite time, but $\psi(p) < 0$ for some medial limit function ψ . Then $p \in \mathcal{R}_X^g$: reconstructed, but only by centres p ’s own ascent cannot find — a remote ball, or a drifting sequence.*
- (3) (Escaping.) *Neither. Then $p \notin \mathcal{R}_X^g$, and every capture curve from p escapes to infinity with $d_X(\Phi_t) = d_X(p) + t$.*

In particular $\mathcal{R}_X^g = \{p : (1) \text{ or } (2)\}$, and on compact M class (1) is all of $M \setminus X$.

Proof. The classes are exclusive and exhaustive by construction. (1) $\Rightarrow p \in \mathcal{R}_X^g$ is Theorem 3.1(i); (2) $\Rightarrow p \in \mathcal{R}_X^g$ is Lemma 4.1. Conversely, if $p \in \mathcal{R}_X^g$ then $\psi_a(p) < 0$ for the finite medial limit function of some containing ball, so p lies in class (2) whenever not in class (1) — never in (3). A point outside class (1) has every capture curve uncaptured, hence escaping, by Theorem 3.1. $\square$

Białozyt’s theorem is not a description of the reconstructed set but a *reconstructibility criterion*: it says exactly when a closed set is recovered in full. The trichotomy yields the direct analogue on any complete manifold, because $\mathcal{R}_X^g$ is precisely classes (1)–(2) while class (3) is its complement.

Corollary 4.5 (Reconstructibility criterion). *Let M be complete, X closed, $\emptyset \neq X \neq M$. The following are equivalent.*

- (a) *X is reconstructible: $\mathcal{R}_X^g = M \setminus X$.*

- (b) $M \setminus X \subseteq \bigcup \{\psi < 0\} : \psi \text{ a medial limit function of } X\}.$
- (c) *Every $p \in M \setminus X$ is captured or drift-swallowed; equivalently, the escaping class (3) of Theorem 4.4 is empty.*
- (d) *For every $p \in M \setminus X$ whose capture curves escape to infinity, some medial limit function ψ satisfies $\psi(p) < 0$.*

Proof. (a) $\Leftrightarrow$ (b): by Theorem 4.3 the right-hand side of (b) equals $\mathcal{R}_X^g$, and $\mathcal{R}_X^g \subseteq M \setminus X$ always, so (b) is the reverse inclusion, i.e. (a). (a) $\Leftrightarrow$ (c): by Theorem 4.4, $\mathcal{R}_X^g$ is exactly classes (1)–(2) and is disjoint from class (3); since the three classes partition $M \setminus X$, $\mathcal{R}_X^g = M \setminus X$ iff class (3) is empty. (c) $\Leftrightarrow$ (d): a point outside class (1) escapes (Theorem 3.1), so it lies in class (2) — swallowed — iff some ψ has $\psi(p) < 0$; captured points lie in $\mathcal{R}_X^g$ regardless, so emptiness of class (3) is exactly (d). $\square$

Two remarks fix its meaning. First, *on a compact manifold the criterion is vacuously satisfied*: escape is impossible (Theorem 3.1), class (3) is empty, and X is reconstructible for free (Corollary 3.2). The criterion therefore has content only on non-compact M — exactly where Białożyty’s does, $\mathbb{R}^n$ being non-compact. Second, condition (b) is the *unconditional* form: it refers to the medial limit functions, hence implicitly to the axis M_X^g . Białożyty’s theorem is the *faithful* sharpening of (b) in the flat case, replacing “ $M \setminus X \subseteq \bigcup \{\psi < 0\}$ ” by the axis-free hull condition

$$\mathbb{R}^n \setminus X \subseteq \overline{\text{conv}} X \cup \bigcup_{L \text{ defective}} L^+$$

of Theorem 1.1 (Corollary 5.1); on a Hadamard manifold the corresponding faithful sharpening replaces the ideal sublevel sets by defective horoballs $\{b_\xi < -c(\xi)\}$ (§5.3, Remark 4.7). Thus Corollary 4.5 is the general analogue of Białożyty’s main theorem, and the passage from (b) to a checkable-from- X certificate is precisely the faithful-layer problem, solved in flat and Hadamard geometry and open in mixed curvature.

Remark 4.6 (The witnesses of class (2)). Both kinds of witness genuinely occur, already for $X = \{x_0, x_1\} \subseteq \mathbb{R}^2$ and p beyond x_0 in an open half-plane at an obtuse angle to the bisector. Every capture curve from p runs straight to infinity with the single foot x_0 , yet p lies in a single *finite* bisector ball (and in the ideal half-plane limit of the bisector balls). Class (2) is thus the precise sense in which “reconstruction is non-local for p ”: the certificate cannot be found by ascending from p . This is exactly the phenomenon that separates the non-compact theory from the compact one, and the reason the faithful certificates below must quantify over supporting half-spaces or horoballs rather than over flow lines.

Remark 4.7 (The Hadamard offset, dissolved). Let M be Hadamard (complete, simply connected, $\text{sec} \leq 0$). By Remark 2.6 the ideal medial limit functions are exactly $b_\xi + c$ with $\xi \in \partial_\infty M$ and $c \in \mathbb{R}$, normalised so that b_ξ vanishes on the horosphere of the maximal free horoball $\mathcal{H}$ at ξ ; necessarily $c \geq 0$ when X meets every neighbourhood of that horosphere. Call $\mathcal{H}$ *defective at level c* if $b_\xi + c$ is a medial limit function; then Lemma 4.1 reconstructs $\{b_\xi < -c\}$, the horoball shrunk by c . The historical “radial offset” $\text{off}_k = d(o, a_k) + b_\xi(a_k)$ is precisely the discrepancy $\psi_{a_k} - b_\xi$ along the drift, so a drift with $\text{off}_k \rightarrow c$ is a level- c certificate. What used to be a gap — “only the deep part $\{b_\xi < -c\}$ is reconstructed for a non-vanishing asymptotic offset” — is now simply the content of the level: the drift *is* a level- c certificate and reconstructs

exactly $\{b_\xi < -c\}$. The full-horoball statement is the assertion that a level-0 certificate exists, and the exact level

$$c(\xi) := \inf\{c \geq 0 : b_\xi + c \text{ is a medial limit function}\}$$

is a question about X and the horosphere geometry, not about the medial axis; see §7.

5 Specialisations and worked examples

We now recover the classical theory as specialisations of Theorems 4.3–4.4, and illustrate every key definition with an explicit example. Throughout, the *unconditional* content — the abstract characterisation — is supplied by the theorems above; the *faithful* content — a certificate readable from X and M — is supplied by Białożyty’s theorem in the flat case and discussed as the open residue elsewhere.

5.1 Euclidean space: recovering Białożyty’s theorem

In $\mathbb{R}^n$ geodesics are unique, so $M_X^g = M_X =: M_X$ and $\mathcal{R}_X^g = \mathcal{R}_X =: \mathcal{R}_X$. We identify the medial limit functions. A finite medial limit function is a free-ball function $y \mapsto |y - a| - r$, $a \in \overline{M_X}$. For an *ideal* one, let $a_k \rightarrow \infty$ along a unit direction u (i.e. $a_k/|a_k| \rightarrow u$) with reaches $r_k = d_X(a_k)$. Then, for each fixed y ,

$$\psi_{a_k}(y) = |y - a_k| - r_k = (|y - a_k| - |a_k|) + (|a_k| - r_k) \longrightarrow -\langle y, u \rangle + \beta,$$

where the first bracket tends to $-\langle y, u \rangle$ (a standard Euclidean computation) and convergence of ψ_{a_k} forces the offset $|a_k| - r_k$ to a finite limit β . So every ideal medial limit function is *affine*,

$$\psi(y) = \beta - \langle y, u \rangle, \quad \{\psi < 0\} = \{\langle y, u \rangle > \beta\},$$

an open half-space with outward normal u . Conversely a supporting half-space limit of medial balls is exactly such a ψ . Theorem 4.3 therefore reads, in $\mathbb{R}^n$:

$$\mathcal{R}_X = \left(\bigcup \text{finite medial balls} \right) \cup \left(\bigcup \{ \text{open half-spaces that are ideal medial limits} \} \right).$$

This is Białożyty’s decomposition. The finite balls fill $\text{conv } X \setminus X$ (every point of the convex hull lies in a free ball); the half-spaces are the outer half-spaces L^+ of supporting hyperplanes L that arise as ideal limits. Białożyty’s Theorem 1.1 identifies *which* supporting hyperplanes occur: precisely the *defective* ones, $X \cap L \neq \overline{\text{conv}} X \cap L$ (equivalently, by Motzkin’s theorem, those carrying a non-Chebyshev point just outside the contact face). Thus:

Corollary 5.1 (Flat specialisation). *In $\mathbb{R}^n$, Theorem 4.3 together with Białożyty’s Theorem 1.1 gives*

$$\mathcal{R}_X = (\overline{\text{conv}} X \setminus X) \cup \bigcup_{L \text{ defective}} L^+,$$

and X is reconstructible iff $\mathbb{R}^n \setminus X \subseteq \overline{\text{conv}} X \cup \bigcup_L L^+$. The drift over a defective L runs at level $\beta = 0$ (no offset): the escaping bisector centres accumulate their feet on the defective contact face, and their half-space limit is L^+ itself.

Here the split of labour is exact and instructive: the abstract characterisation (unconditional, self-contained) delivers “ $\mathcal{R}_X$ is balls $\cup$ half-space limits”; Białożyty’s theorem (the faithful certificate) delivers “the half-space limits are exactly the defective supporting hyperplanes.” This is the template the whole paper follows on other geometries.

Example 5.2 (Two points). $X = \{x_0, x_1\} \subseteq \mathbb{R}^2$, so $\overline{\text{conv}} X$ is the segment $[x_0, x_1]$ and the axis M_X is its perpendicular bisector. There is exactly one defective supporting line: the line ℓ through x_0x_1 itself, on which $\overline{\text{conv}} X \cap \ell = [x_0, x_1]$ while $X \cap \ell = \{x_0, x_1\}$ — two points, not the whole segment. It supports from both sides, so it is defective in both orientations, with the two open half-planes bounded by ℓ as outer half-spaces L^+ ; these are the ideal (half-plane) medial limits of the escaping bisector balls. By Corollary 5.1,

$$\mathcal{R}_X = ([x_0, x_1] \setminus \{x_0, x_1\}) \cup \{\text{two open half-planes}\} = \mathbb{R}^2 \setminus (\ell \setminus (x_0, x_1)),$$

i.e. *everything except the two outer rays* of ℓ beyond the endpoints (together with X itself). Note what is and is not reconstructed: the *open* segment (x_0, x_1) is reconstructed — by finite bisector balls, which contain every interior segment point — while the two collinear rays $\{x_0 + t(x_1 - x_0) : t > 0\}$ and its mirror are on *no* free ball and are missed by the open half-planes. Removing the reconstruction of these rays is exactly the failure of reconstructibility for a two-point set. This is the primal drift example of Remark 4.6: a point in an open half-plane at an obtuse angle to the bisector is reconstructed only by the ideal half-plane limit (class (2)), not by its own escaping capture curve.

Example 5.3 (A triangle: the outer half-spaces close up). Let X be the three vertices of an equilateral triangle, so $\overline{\text{conv}} X$ is the solid triangle. A supporting line through a single vertex has $X \cap L = \overline{\text{conv}} X \cap L = \{v\}$ — not defective. A supporting line *along an edge* has $\overline{\text{conv}} X \cap L$ the whole edge while $X \cap L$ is its two endpoints — defective, with a genuinely nonempty outer half-plane L^+ reconstructed by the ideal drift of the escaping bisector centres of those two vertices (which are legitimate medial centres far out beyond the edge, the third vertex being strictly farther). The three edge half-planes *overlap to cover every exterior point* — a point beyond a vertex lies in the outer half-plane of an incident edge — and the interior is filled by finite balls, so $\mathcal{R}_X = \mathbb{R}^2 \setminus X$: reconstructible. Contrast Example 5.2, the degenerate “triangle” with one edge: there the single edge’s two outer half-planes leave the collinear rays uncovered, and reconstructibility fails. The moral is that reconstructibility is about whether the defective outer half-spaces *tile* the exterior, which a genuine polygon does and a segment does not.

5.2 Compact manifolds, the sphere, and space forms

Here the faithful layer is *vacuous*: there are no ideal medial limit functions (the manifold is bounded), no hull condition, and Corollary 3.2 gives total reconstruction outright.

Corollary 5.4 (Round sphere and spherical space forms). *For every closed $\emptyset \neq X \neq \mathbb{S}^n$, $\mathcal{R}_X^g = \mathbb{S}^n \setminus X$; likewise $\mathcal{R}_X^g = (\mathbb{S}^n/\Gamma) \setminus X$ on every spherical space form. For $|X| \geq 2$ under the spread hypothesis $d_X < \text{inj}$ one also gets $\mathcal{R}_X = \mathbb{S}^n \setminus X$ (Corollary 3.3).*

Example 5.5 (A single point on $\mathbb{S}^n$: the antipodal centre). $X = \{x_0\} \subseteq \mathbb{S}^n$. Then $M_X = \emptyset$, so the two-point theory reconstructs nothing. But the antipode $\bar{x}_0$ is a medial centre in M_X^g : it is joined to x_0 by an $(n-1)$ -sphere of minimising geodesics, all of length $\pi = d_X(\bar{x}_0)$, and the free ball $B(\bar{x}_0, \pi) = \mathbb{S}^n \setminus \{x_0\}$ swallows everything. Corollary 5.4 thus reconstructs the whole complement from a single point — the extreme demonstration that M_X^g , not M_X , is the

reconstruction axis. The capture curve from any $p \neq x_0, \bar{x}_0$ ascends the meridian away from x_0 and reaches $\bar{x}_0$ in time $\pi - d_X(p)$: capture, class (1).

Example 5.6 (Cut-shielding and the two-point axis). On a prolate ellipsoid of revolution, or on $\mathbb{S}^n$ with a sparse X , M_X^g can strictly contain M_X : a capturing centre a may have two minimising geodesics to a *single* foot (a cut point of a before any second foot appears), so $a \in M_X^g \setminus M_X$ and the point it captures is reconstructed by $\mathcal{R}_X^g$ but not by $\mathcal{R}_X$. This is the *cut-shielding* obstruction, and it is why the two-point theorem needs a spread hypothesis (Corollary 3.3) while the two-geodesic theorem never does. The obstruction is quantitative: on prolate ellipsoids it switches on past an eccentricity threshold (numerically near $c^* \approx 2.6$; see §7). Class-(1) points of $\mathcal{R}_X^g$ whose capturing centres lie in $M_X^g \setminus M_X$ are exactly the shielded points.

5.3 Hyperbolic space and Hadamard manifolds

Let M be Hadamard. The ideal medial limit functions are Busemann functions at a level (Remark 4.7), so Theorem 4.3 reads: $\mathcal{R}_X^g$ is the union of the finite medial balls and the shrunk horoballs $\{b_\xi < -c(\xi)\}$ over the defective ideal points ξ . This is the exact Hadamard analogue of Corollary 5.1, with “half-space” replaced by “horoball” and “conv” by the horoconvex hull.

The *faithful* question — which ideal points ξ are defective, and at what level — is the curved analogue of Białożyty’s defective-hyperplane certificate. In real hyperbolic space and, more generally, on Hadamard manifolds admitting a faithful horoconvexity certificate, the answer mirrors the flat one: ξ is defective iff the contact set $X \cap \partial\mathcal{H}$ on the maximal free horosphere fails to be horoconvex, i.e. $X \cap \partial\mathcal{H} \neq \text{hconv}(X \cap \partial\mathcal{H})$, and the level is then 0. We record this as the natural specialisation; its proof is the horoconvex counterpart of Motzkin’s theorem and is beyond the flat certificate we assume here (see §7 and the discussion in Remark 4.7).

Example 5.7 (Two ideal directions in $\mathbb{H}^2$). Let $X = \{x_0, x_1\}$ in the hyperbolic plane. The bisector is a complete geodesic with two ideal endpoints $\xi_\pm \in \partial_\infty \mathbb{H}^2$. Escaping bisector centres $a_k \rightarrow \xi_\pm$ have reaches $r_k = d(a_k, x_0) \rightarrow \infty$, and $\psi_{a_k} = d(\cdot, a_k) - r_k \rightarrow b_{\xi_\pm}$, a Busemann function at level 0; by the swallow lemma the two open horoballs $\{b_{\xi_\pm} < 0\}$ are reconstructed. Together with the finite bisector balls, which fill a lens-shaped region around the geodesic segment $[x_0, x_1]$, these exhaust $\mathcal{R}_X^g$: by Theorem 4.3 every reconstructed point lies in a finite lens ball or in one of the two horoballs. This is the hyperbolic mirror of Example 5.2, with “open half-plane” replaced by “open horoball” — the escaping centres now run to the two *ideal points* of the bisector rather than off to Euclidean infinity, and the swallowed regions are horoballs there. The contact set on the maximal free horosphere at each $\xi_\pm$ is the single pair $\{x_0, x_1\}$, non-horoconvex, which is the horoconvex analogue of the defective supporting line and certifies the defect. Whether the two horoballs and the lens leave any point uncovered — the hyperbolic counterpart of the two collinear outer rays — depends on the precise horoball geometry and is a faithful-layer question (§7); the swallow and characterisation above are unconditional.

Example 5.8 (The half-space limit is genuinely ideal). In both Examples 5.2 and 5.7 the drifted region is reconstructed by *no* finite ball: a point p deep in the open half-plane / horoball, at an obtuse angle to the bisector, has its capture curve escaping with a single foot (Remark 4.6), so it is class (2), reconstructed only by the ideal limit. This is why Theorem 4.3 must include the ideal members and why the pure capture theorem (Corollary 3.2) cannot hold on a non-compact manifold.

5.4 Non-compact positive curvature

Here reconstruction is genuinely partial and neither the flat nor the Hadamard certificate applies; the trichotomy still governs.

Example 5.9 (Paraboloid of revolution). Let M be a paraboloid of revolution ($\text{sec} > 0$, a single flattening end) and X a small circle about the vertex. The vertex is a finite medial centre reconstructing the inner disk: class (1). Every point on the flaring side ascends its meridian forever with a single foot, and no medial limit function reaches it: class (3). Hence $\mathcal{R}_X^g \subsetneq M \setminus X$ — partial reconstruction, decided by the non-compactness of M , not by the sign of the curvature.

Example 5.10 (Ideal centres in positive curvature). Take $X = \{x_0, x_1\}$ on the paraboloid. The bisector is unbounded, and its escaping centres furnish genuinely *ideal* medial limit functions on a $\text{sec} > 0$ manifold — consistent with the trichotomy, because their sublevel sets are unbounded *non-convex* free regions. It is the maximal free *convex* regions that are bounded in positive curvature, not the carriers of medial limit functions; the distinction matters, and the example shows the ideal class is not a nonpositive-curvature artefact.

5.5 The completed picture

Two independent axes organise the theory. *Compactness* decides partial versus total; the *sign of the Busemann/Hessian data* decides the flavour of the faithful certificate. The following table records the two-geodesic theory, with the two-point refinement in brackets.

	$\text{sec} \leq 0$	$\text{sec} \equiv 0$	$\text{sec} > 0$
non-compact	partial; hconv defect at level $c(\xi)$	partial; conv defect (Białożyty [1])	partial (paraboloid) finite + ideal centres
compact	$\text{total } \mathcal{R}_X^g$ [M_X : shielding]	$\text{total } \mathcal{R}_X^g$ [M_X : shielding]	$\text{total } \mathcal{R}_X^g$ [M_X : shielding; = $\mathbb{S}^n$ ok]

The compact row is Corollary 3.2 — unconditional, all curvatures, $|X| \geq 1$. The non-compact row is Theorem 4.3, with the faithful certificate known on the flat column (Białożyty) and its Hadamard extension, and open in mixed curvature (§7).

6 Discussion: the two layers

It is worth stating plainly what has been achieved and what the shape of the theory is, because the two are easy to conflate.

The unconditional layer is complete. On every complete Riemannian manifold and for every nonempty closed proper X , the reconstruction $\mathcal{R}_X^g$ is exactly the union of the sublevel sets of the medial limit functions (Theorem 4.3); every point is captured, drift-swallowed, or escaping (Theorem 4.4); and reconstruction is total on every compact manifold (Corollary 3.2). No curvature bound, genericity, analyticity, spread, or cardinality hypothesis appears anywhere in this layer. The two ideas that make it work are worth isolating:

- *Stop at the axis, do not flow through it.* The margin one would like to accumulate is not monotone through multi-geodesic points (Remark 3.4); but the axis is the *goal*, so stopping at its first approach both avoids the non-monotonicity and delivers capture. This is what

removes the dependence, in every previous compact argument, on an unproven extension of the swallow property past cut points.

- *Normalise limits by the reach.* Measuring a free ball by its depth $d_X(a)$ rather than by the distance to a fixed basepoint makes the limit objects basepoint-free and absorbs the historical “offset” into the function itself (Remarks 2.6, 4.7). The swallow lemma then costs three lines on *any* complete manifold.

The faithful layer is curvature-specific and only partly known. Replacing “ ψ is a medial limit function” by a certificate readable from X and the geometry of M is exactly the content of Białożył’s theorem in the flat case (Corollary 5.1), of the horoconvex-defect certificate on Hadamard manifolds (§5.3), and it is vacuous on compact manifolds. In mixed curvature it is open (§7). The honest summary is: the *dynamics* of reconstruction are understood in full generality; the *combinatorics* of which asymptotic directions carry a defect are understood exactly where a Motzkin-type theorem is available, and Białożył’s theorem is the flat instance of such a theorem.

7 What remains

The open problems live entirely on the faithful (escape) side.

- (1) *Certificates in mixed curvature.* For which ideal points ξ (horofunctions h) of a general non-compact M is $h + c$ a medial limit function, read from the contact structure of X on $\{h = 0\}$? Known: the conv/Motzkin certificate in $\mathbb{R}^n$ (Białożył [1], Motzkin [8]); the horoconvex certificate on Hadamard manifolds. Unknown already for a complete surface carrying both signs of curvature.
- (2) *The level function.* On an inhomogeneous Hadamard manifold, characterise $c(\xi) = \inf\{c : b_\xi + c \text{ a medial limit}\}$ from X alone (Remark 4.7). One has $c(\xi) = 0$ under curvature pinching, in the flat case, and over any bisector witness on the horosphere, and no example with $c(\xi) > 0$ is known; deciding whether the deep-part phenomenon occurs at all is the sharpest open question on the Hadamard side.
- (3) *The two-point axis in variable curvature.* Quantify cut-shielding (Example 5.6): the prolate-ellipsoid threshold $c^* \approx 2.6$ awaits an analytic determination, as does the class of compact M with $\mathcal{R}_X = M \setminus X$ for all sparse X (constant curvature suffices; bounded eccentricity conjecturally).
- (4) *Chebyshev-type problems on the frontier.* The Motzkin question in Carnot groups (e.g. $\mathbb{C}\mathbb{H}^n$ horospheres) and in Alexandrov spaces of indefinite curvature (pinched horospheres); and the domain (Lieutier) problem on $\mathbb{S}^n$, where the boundary’s own positively curved intrinsic geometry becomes the detection object.

Conclusion. Białożył’s Theorem 6 generalises to every complete Riemannian manifold, in two layers. The *unconditional* layer, proved here in full from that theorem alone, says that reconstruction by the two-geodesic axis is exactly the union of the sublevel sets of the medial limit functions (Theorem 4.3); that every point is captured, swallowed, or escaping (Theorem 4.4); and that capture is total on every compact manifold (Corollary 3.2), with no hypothesis beyond

completeness and closedness. The *faithful* layer — replacing “is a medial limit function” by a certificate on contact sets — is exact on the flat knife-edge (conv, via Białożyty), on Hadamard manifolds (hconv), and trivial on compact manifolds; its extension into mixed curvature, and the Chebyshev-set problems it generates, are the genuine residue. The two analytic obstructions that historically blocked the Riemannian passage — the containment step past cut points and the radial offset — are removed here, the first by stopping the ascent at its first approach to the axis instead of flowing through it, the second by normalising the limit objects by their own reach.

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